

PONGANGAIRAMAN NKS

Senior Frontend Engineer · UI Architect · React/TypeScript, Next.js

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KEY ACHIEVEMENTS

5M+ Learners Dashboard Revamp	75% Faster Renders Virtualization & Optimization	500K+ Msgs/Month Campaign Engine	50K+ Active Users Real-time Community	INR 2Cr+ GMV/Month Analytics Dashboard
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PROFILE SUMMARY

Senior Frontend Engineer with **4+ years of experience** specializing in building high-concurrency, **scalable web architectures for 5M+ users** at **Unacademy Group**. Expert in React, TypeScript, and Next.js, with a proven track record of **reducing TTI by 75%** and architecting real-time systems supporting **10,000+ concurrent viewers**. Recognized for bridging the gap between complex system design and pixel-perfect user experiences, driving creator retention and engagement at an enterprise scale.

PROFESSIONAL EXPERIENCE

Frontend Engineer | Unacademy – Graphy Labs Jun 2025 – Present

Architecting High-Concurrency Product Suites for 5M+ Global Learners & Creators

- Architected a high-concurrency real-time community platform using **React and Zustand**, maintaining **<100ms message delivery** for **50k+ active members**.
- Led the architectural overhaul of the **Learner Dashboard for 5M+ users**; implemented virtualization and optimized hydration cycles to **reduce TTI from 3.2s to 800ms**.
- Designed and built a comprehensive **Analytics Dashboard** using Chart.js with real-time Line, Bar, and Area charts, tracking total users, transactions (**INR 2Cr+ monthly GMV**), and engagement metrics across **1,000+ communities**.
- Engineered a live-streaming interface for **10,000+ concurrent viewers**, utilizing Cloudflare Realtime Kit to achieve stream **initialization in <200ms**.
- Developed an enterprise campaign engine processing **500k+ monthly messages**, featuring a custom drag-and-drop builder that increased creator **onboarding efficiency by 70%**.
- Established a real-time **Telegram-to-Web sync pipeline** handling **100k+ daily events**, enhancing engagement for the platform's **top 500 creators**.
- Improved platform **stability by 30%** by implementing automated integration testing and custom **Sentry error-tracking middleware**.

Software Engineer II | Mirafra Technologies Jan 2025 – Jun 2025

Leading Modular UI Development for Enterprise-Grade OTT SaaS Solutions

- Architected the frontend of a **white-label OTT SaaS platform** using Next.js and SSR, achieving a **sub-300ms Time to First Byte (TTFB)**.
- Implemented a robust security layer including **JWT-based authentication, RBAC**, and secure session persistence **across multiple subdomains**.
- Automated multi-stage **CI/CD pipelines** via GitHub Actions, reducing production **deployment time by 75%**.

React/TypeScript Developer | Dataphi Analytics Labs Jun 2022 – Dec 2024

Developing Data-Intensive Recruitment & FinTech Ecosystems for Enterprise Clients

- Built a **multi-tenant recruitment platform for 50+ enterprise firms**, featuring a Kanban-based workflow that reduced average “time-to-hire” by **30%**.
- Engineered a **unified UI component library** adopted across 3 flagship products, **reducing development cycles by 35%** and ensuring brand consistency.
- Developed a high-traffic **gaming CMS managing INR 50L+** in distributions; optimized admin workflows by **80%** through advanced table virtualization and caching.

Frontend Developer | Krayen (Full Time – Intern) Feb 2022 – Jun 2022

Founding Engineer: Scaling Frontend Infrastructure from MVP to Seed Round

- Founded the initial frontend architecture**, implementing Redux state management and a modular structure that scaled through the company’s seed round.
- Integrated **Firebase** for real-time data persistence and authentication, delivering a functional MVP within a strict **3-month timeline**.

PROJECTS

- NeuroBot**: Architected a visual node-based editor with **15+ complex node types** and a **RAG pipeline** utilizing semantic caching to **reduce LLM API costs by 40%**. Published a production-ready, themeable, embeddable **SDK supporting ESM, CJS, and UMD formats** with zero global CSS leakage.
- IDColab**: Engineered an enterprise file viewer **supporting Three.js 3D models, 360° panoramas, and PDF annotations** for design assets exceeding 50MB. Implemented interactive Shadow Study tools and progressive loading to maintain high-performance rendering during spatial analysis.
- AI Resume Automation**: Developed an **LLM-powered platform to generate ATS-optimized (LaTeX) resumes**, reducing tailoring time from 45 minutes to 3 minutes.

CORE COMPETENCIES

Frontend Engineering: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Responsive Web Design, Single Page Applications (SPA), Server-Side Rendering (SSR), Static Site Generation (SSG)

State Management & Data Layer: Redux (Toolkit), Zustand, React Query (TanStack Query), Context API, GraphQL (Apollo Client), REST APIs, Axios, WebSocket/Real-time Data Handling

Performance & Architecture: Micro-frontend Architecture, Code Splitting, Lazy Loading, Tree Shaking, Virtualization (react-window/react-virtualized), Bundle Optimization (Webpack/Vite), Performance Profiling (Lighthouse, Chrome DevTools), Core Web Vitals Optimization, PWA, Memoization Strategies

UI & Design Systems: Material UI (MUI), Tailwind CSS, Bootstrap, Custom Component Libraries, Design System Architecture, Accessibility (a11y, WCAG 2.1), CSS-in-JS (Styled Components/Emotion)

Real-time, Integrations & Mobile: WebSockets, Webhooks, Live Streaming (Cloudflare Realtime Kit), Payment Gateways (Razorpay, Stripe, Chargebee), React Native

Tools, Testing & DevOps: Webpack, Vite, Git, CI/CD (GitHub Actions), Jest, React Testing Library, Sentry, Google Analytics, Docker, Vercel

Monitoring & Observability: Sentry (Error Tracking), Google Analytics, Mixpanel/Amplitude (Event Tracking), Performance Monitoring

EDUCATION

B.E | Govt. College of Technology, Coimbatore

2017 – 2021 | **CGPA: 8**